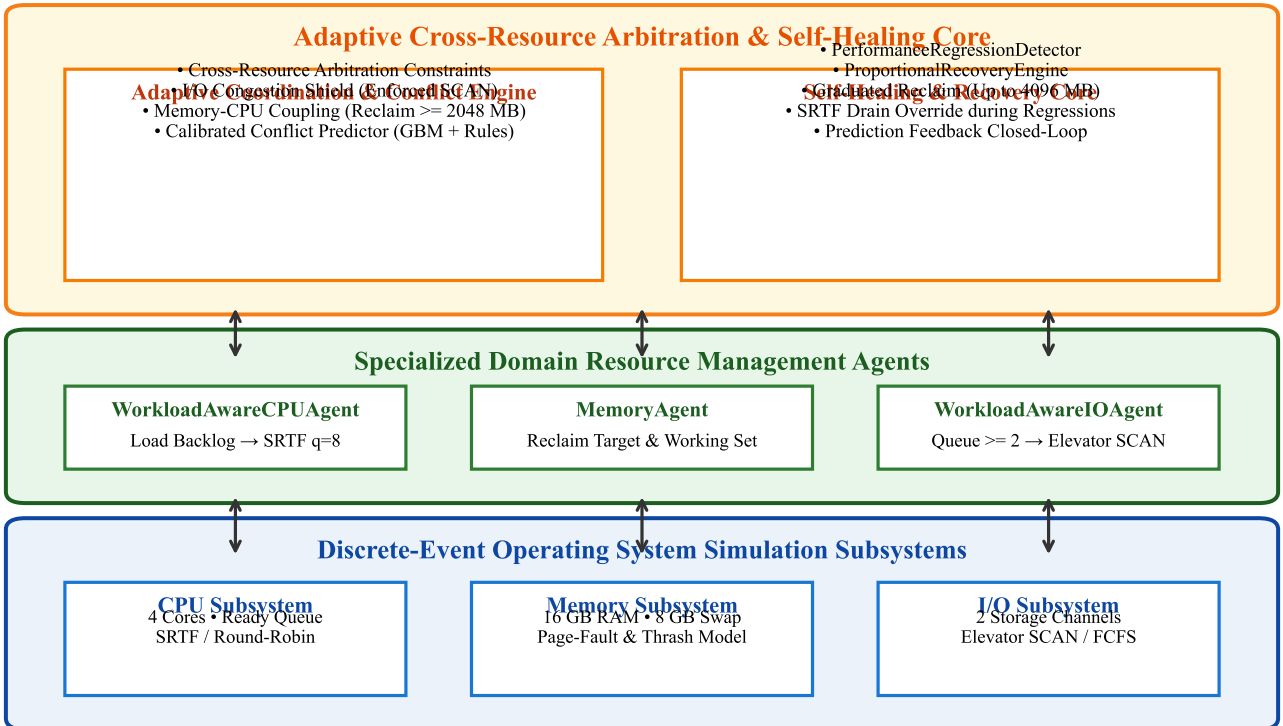
Adaptive Cross-Resource Arbitration & Self-Healing Core
Adaptive Coordination & Conflict Engine
• Cross-Resource Arbitration Constraints
• I/O Congestion Shield (Enforced SCAN)
• Memory-CPU Coupling (Reclaim >= 2048 MB)
• Calibrated Conflict Predictor (GBM + Rules)
Self-Healing & Recovery Core
• PerformanceRegressionDetector
• ProportionalRecoveryEngine
• Graduated Reclaim (Up to 4096 MB)
• SRTF Drain Override during Regressions
• Prediction Feedback Closed-Loop
Specialized Domain Resource Management Agents
WorkloadAwareCPUAgent
Load Backlog → SRTF q=8
MemoryAgent
Reclaim Target & Working Set
WorkloadAwareIOAgent
Queue >= 2 → Elevator SCAN
Discrete-Event Operating System Simulation Subsystems
CPU Subsystem
4 Cores • Ready Queue
SRTF / Round-Robin
Memory Subsystem
16 GB RAM • 8 GB Swap
Page-Fault & Thrash Model
I/O Subsystem
2 Storage Channels
Elevator SCAN / FCFS